Towards Accurate Ultrasound Nerve Segmentation via Deep Learning

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Summary: This work explores different deep learning approaches for segmenting the brachial plexus nerve in ultrasound images to address challenges like low contrast, speckle noise, and ambiguous boundaries. Three architectures are evaluated. Vanilla U-Net as out baseline model achieved a test Dice score of 0.76 on raw data, demonstrating robustness across preprocessing methods like 2D filtering and CLAHE. Then, we enhanced the model with multi-scale inception modules and CLAHE preprocessing, this model achieved a test Dice score of 0.81 after integrating OneCycleLR scheduling to reduce overfitting and improving generalization. Moreover, a transformer-based model (SegFormer) was implemented by using hierarchical attention and an MLP decoder. With CLAHE preprocessing this model achieved a testing Dice score of 0.75 when using testing images of raw dataset. CLAHE improved performance for both Inception U-Net and SegFormer, while 2D filtering offered marginal gains. Future work could focus on hybrid architectures and improved preprocessing.

Related work: Ultrasound is widely used in clinical setting for its non-invasive nature, real-time capability, and absence of radiation exposure. As such, ultrasound nerve segmentation can become a vital task for procedures such as regional anesthesia and the diagnosis of neuropathy. Nerve segmentation in ultrasound provide unique difficulties as they often appear as small with indistinct boundaries which will complicate the differentiation from tendons and artifacts. Traditional methods for nerve segmentation relied on handcrafted features and classical image processing techniques such as edge detection and thresholding [8]. However, the inherent challenges of ultrasound like low contrast, speckle noise, and anatomical variability have inspired many to do research on automated segmentation methods [6]. In 2015, Ronneberger *et al.* [9] introduced the U-Net architecture and soon became a foundational architecture for medical image segmentation due to its effectiveness in segmenting biomedical images with limited training data where annotated datasets are often scarce. This architecture has a symmetric encoder-decoder design with skip connections that enables precise localization by combining high-resolution contextual information from the encoder with upsampled feature maps in the decoder. U-Net has shown success in organ segmentation to semi-automatically and fully automatically segment a 3D volume from a sparse annotation and has offered an accurate segmentation for highly variable structure of the of the Xenopus kidney [1]. To address U-Net's limitations, subsequent studies introduced some architectural modification. He *et al.* introduced residual connections [3] and it got integrated into U-Net variants to stabilize gradient flow and enable deeper network [12]. To overcome noisy ultrasound data, attention mechanisms like squeeze-and-extraction blocks proposed by Hu *et al.* [4] was used to prioritize salient features [10]. "Improved U-Net Model for Nerve Segmentation" [13] adopts inception modules and batch normalization to achieve fewer parameters and less time for training. In addition, they used Dice coefficient as loss function that compares the pixel-wise agreement with a predicted segmentation and ground truth (Fig. 1).

Figure 1. From left to right: the raw ultrasound image containing nerve structure, mask manually annotated by experts, and ultrasound image with the ground truth segmentation map (red border). This figure is from [13]

Methods/Networks:

Preprocessing with CLAHE and 2D Filtering.

To enhance image quality and reduce noise, we applied two common preprocessing techniques: CLAHE and 2D filtering. CLAHE (Contrast Limited Adaptive Histogram Equalization) improves local contrast in ultrasound images by adaptively redistributing pixel intensities in small regions, which is especially helpful in visualizing subtle anatomical boundaries like nerves. However, CLAHE can also amplify noise. To address this, we implemented a 2D filtering step using a 3×3 convolutional kernel. This operation computes the new value of each pixel as a weighted sum of its 3×3 neighborhood and effectively smooths high-frequency artifacts. The filtered center pixel is calculated as:

$$R_5 = \sum_{i=1}^{9} R_i G_i \tag{1}$$

where R is the local image region and G is the filter kernel. In our implementation, we use a uniform averaging kernel (each $G_i = \frac{1}{9}$), which suppresses speckle noise while preserving the overall nerve structure. These preprocessing steps were applied individually and in combination across the dataset to study their effect on model performance.

Vanilla U-Net:

To establish a strong and interpretable baseline, we implemented the original U-Net architecture proposed by Ronneberger *et al.* [9] for the task of ultrasound nerve segmentation. The vanilla U-Net is a fully convolution network that follows a symmetric encoder-decoder architecture, with skip connections that preserve spatial context by forwarding feature maps from the encoder to the corresponding decoder stages. This architecture is particularly well-suited for biomedical image segmentation tasks where precise localization is necessary despite limited annotated data.

The encoder consists of four levels of convolution blocks. Each block contains two successive 3×3 convolution layers, each followed by batch normalization and ReLU activation. Max pooling layers with a stride of 2 are inserted after each block to progressively reduce spatial resolution while increasing the feature depth. At the bottleneck of the network, a final convolution block operates at the lowest resolution with the largest receptive field.

The decoder path mirrors the encoder. Each decoder level begins with a transposed convolution to upsample the feature maps. These are concatenated with the corresponding encoder features via skip connections, ensuring that fine-grained spatial details are preserved. The concatenated features are passed through convolution blocks similar to those in the encoder. A final 1×1 convolution maps the output to a single channel followed by a sigmoid activation, producing the final probability mask.

The entire network was trained from scratch with Xavier (Glorot) initialization for convolution weights. The model was implemented in PyTorch. The following table outlines the architecture of the vanilla U-Net used in our experiments:

Inception U-Net

This work adopts the improved U-Net architecture proposed by Zhao *et al.* [13] that enhances the classical U-Net [9] design by adding inception modules, batch normalization, and optimized upsampling strategies. The goal is to improve nerve segmentation accuracy in ultrasound images by capturing multi-scale features. The model follows a symmetric encoder-decoder structure with skip connections as can be seen in Fig. 2.

In the contracting path, i.e. encoder, there is a series of convolutional blocks and inception modules to extract features. The convolutional blocks have 3×3 convolutional layer followed by batch normalization (BN) and ReLU activation function to accelerate training and mitigating gradient vanishing (Fig. 3). There are four inception modules in this path after the initial convolutions. Each inception module has 4 parallel paths: two paths with 1×1 convolutions

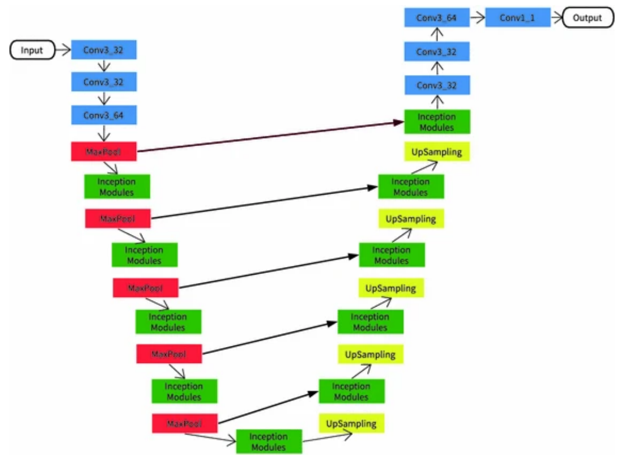

Figure 2. Proposed improved U-Net architecture. Blue box represent basic convolutional units, green box represent inception modules, red box represents downsampling (max pooling), and yellow box is upsampling. This figure is from [13]

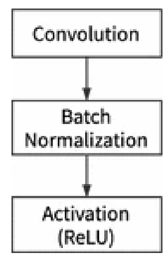

Figure 3. Convolutional block. This figure is from [13]

followed by 3×3 and 5×5 convolutions, respectively. A 1×1 convolutional path for channel reduction and finally, a path with average pooling (that replaces max pooling for noise robustness) followed by a 1×1 convolution. The inception module design can be seen in Fig. 4. By combining 1×1, 3×3, and 5×5 convolutions within a single module, the network captures features at varying receptive fields which is critical for segmenting nerves of different sizes and orientations in ultrasound images. Five max pooling with stride of 2 are in the contracting path to reduce spatial resolution while expanding feature depth.

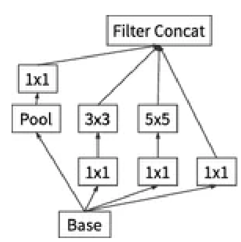

Figure 4. Inception modules with 4 convolutional paths that act on one input and connect to one output with average pooling as pooling layer. This figure is from [13]

After the contracting path, there is a bottleneck layer that

Stage	Layer Description	Output Shape
Input	Input grayscale image	(1, 128, 128)
Encoder 1	2×Conv3×3 + BN + ReLU	(64, 128, 128)
Downsample 1	MaxPool 2×2	(64, 64, 64)
Encoder 2	2×Conv3×3 + BN + ReLU	(128, 64, 64)
Downsample 2	MaxPool 2×2	(128, 32, 32)
Encoder 3	2×Conv3×3 + BN + ReLU	(256, 32, 32)
Downsample 3	MaxPool 2×2	(256, 16, 16)
Encoder 4	2×Conv3×3 + BN + ReLU	(512, 16, 16)
Downsample 4	MaxPool 2×2	(512, 8, 8)
Bottleneck	2×Conv3×3 + BN + ReLU	(1024, 8, 8)
Upsample 1	TransConv 2×2	(512, 16, 16)
Decoder 1	Concat + 2×Conv3×3 + BN + ReLU	(512, 16, 16)
Upsample 2	TransConv 2×2	(256, 32, 32)
Decoder 2	Concat + 2×Conv3×3 + BN + ReLU	(256, 32, 32)
Upsample 3	TransConv 2×2	(128, 64, 64)
Decoder 3	Concat + 2×Conv3×3 + BN + ReLU	(128, 64, 64)
Upsample 4	TransConv 2×2	(64, 128, 128)
Decoder 4	Concat + 2×Conv3×3 + BN + ReLU	(64, 128, 128)
Output	Conv1×1 + Sigmoid	(1, 128, 128)

Table 1. Vanilla U-Net architecture used in this study, showing input/output dimensions at each stage.

is a single inception module to bridge the encoder and decoder and aggregates multi-scale information.

Next, we have an expansive path, i.e. decoder, which is a mirror of the decoder to recover the spatial resolution. This path includes upsampling that is five bilinear interpolation layers that are being used instead of transposed convolutions to reduce artifacts and computational overhead. Then, we have feature fusion that has inception modules and basic convolutional units that process upsampled features. The skip connections in this path concatenate encoder features from the same hierarchical level to preserve fine grained details like nerve boundaries. Finally, a 1×1 convolution with sigmoid activation produces a binary segmentation mask matching the input image dimension.

SegFormer (MiT-B0)

This work adopts the SegFormer architecture proposed by Xie *et al.* [11], a transformer-based segmentation framework that eliminates the need for positional encodings and complex decoder designs. The backbone used is MiT-B0, the smallest variant of the Mix Transformer family, which balances accuracy and efficiency. The goal is to segment nerve structures in ultrasound images through a robust and lightweight model. The model's high-level design includes a hierarchical encoder and an All-MLP decoder, as illustrated in Fig. 5.

In the encoder, the input image is first split into overlapping patches (with kernel size $K = 7$, stride $S = 4$, and padding $P = 3$ for early stages), which are processed through four hierarchical stages. Each stage applies attention blocks with sequence reduction ratios $\{64, 16, 4, 1\}$ respectively to improve computational efficiency. Instead of positional encoding, the encoder uses Mix-FFN blocks with 3×3 depthwise convolutions that inherently capture spatial information, thereby improving flexibility across resolutions.

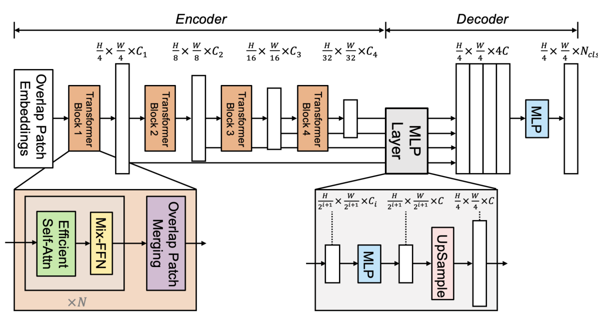

Figure 5. SegFormer architecture. A hierarchical Transformer encoder extracts multiscale features, which are fused by an All-MLP decoder to output semantic segmentation masks. This figure is adapted from [11]

The decoder consists solely of MLP layers. First, features from all encoder stages are projected to a common embedding dimension. These features are then upsampled to a common resolution ($\frac{1}{4}$ of the input size), concatenated, and fused via additional MLP layers to generate the final segmentation mask. The minimal design of the decoder enables efficient training and inference without sacrificing segmentation quality.

Compared to CNN-based designs like DeepLabv3+ or U-Net, SegFormer benefits from a large effective receptive field enabled by non-local Transformer attention. This allows the model to learn both fine-grained and contextual information critical for segmenting nerves of varying sizes and shapes in ultrasound data.

The model was fine-tuned for binary segmentation using a hybrid loss of Binary Cross-Entropy with Logits and Dice loss (foreground only, additional logic gate was added to determine if the foreground is all black), optimized using the Adam optimizer and a One-Cycle learning rate schedule (max LR = 3×10^{-5}). AMP (Automatic Mixed Precision) was used for efficient training. Best checkpoints were selected based on the highest validation Dice coefficient.

Throughout training, we monitored:

- Training and validation losses (BCE + Dice)
- Batch-wise Dice coefficient for both training and validation
- A logic gate to detect and exclude all-black foreground predictions

Final evaluations were conducted using the best checkpoint on all three test splits to assess generalizability.

Datasets, Experiments and Results:
Our dataset is from Ultrasound Nerve Segmentation competition [7]. The task in the competition is to segment a collection of nerves called Brachial Plexus (BP) in ultrasound images. The nerves in the images are manually annotated by humans, and the annotators were trained by experts and instructed to annotate images where they felt confident about the existence of the BP landmark. The dataset contains images where BP is not present. Some images have noise, artifact, and mistakes in the ground truth. There are also identical or very similar images. The competition only provided ground truth masking for training dataset, therefore, we split the training dataset by number of patient into training, validation, and testing dataset with a ratio of 0.7, 0.2, and 0.1 respectively. This training dataset is from 47 patients and in total, with images of 580×420 pixels and their corresponding masks are nerve (white) and background (black).

This task is showing a significant class imbalance, where the nerve region is a small fraction of the total image area compared to the background. Traditional pixel-wise loss functions like binary cross-entropy (BCE) treat all pixels equally that biases the model to the background class and will cause missing sparse nerve structures. The BCE is defined as:

$$\mathcal{L}_{BCE} = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log p_i + (1 - y_i)\log(1 - p_i)], \quad (2)$$

where $y_i \in \{0, 1\}$ is the ground-truth label and $p_i \in [0, 1]$ is the predicted probability for pixel i. Directly optimizing BCE often leads to under-segmentation of nerves, as the model prioritizes background pixel accuracy. To address this issue, we used Dice loss proposed in [13], that directly optimizes the overlap between predicted and ground-truth regions:

$$\mathcal{L}_{Dice} = 1 - \frac{2\sum_{i=1}^{n} y_i p_i}{\sum_{i=1}^{n} y_i + \sum_{i=1}^{n} p_i}. \quad (3)$$

The Dice loss removes the class imbalance by rewarding spatial agreemnet rather than per-pixel accuracy.

Vanilla U-Net Experiments:
We evaluated the vanilla U-Net on three dataset variants: raw, 2D filtered, and CLAHE-enhanced ultrasound images. The model achieved a Dice score of 0.77 on the raw dataset. This result suggests that, despite the presence of speckle noise and variability in image intensity, the vanilla U-Net is capable of effectively capturing the nerve structure. This is likely due to the architectural design that allows for both high-resolution spatial localization and deep contextual feature extraction.

For the 2D filtered dataset, the model achieved a Dice score of 0.76. The marginal drop from the raw dataset indicates that while filtering may reduce some noise, it might also slightly suppress subtle anatomical edges important for identifying nerve boundaries. This reinforces the importance of preserving signal fidelity in preprocessing pipelines.

Similarly, the CLAHE-enhanced dataset yielded a Dice score of 0.76. CLAHE is intended to improve local contrast, especially in homogeneous regions, but its effects appear neutral in this context. The nearly identical performance across all three datasets supports the robustness of the vanilla U-Net architecture to different forms of intensity-based preprocessing.

Table 2 summarizes these test results across the three conditions. The consistent performance across datasets, each representing a different preprocessing strategy, provides evidence that even the unmodified U-Net serves as a reliable segmentation tool in this application.

Filter	Test Dice Score	Test IoU Score
Raw	0.7655	0.6438
2Dfilter	0.7628	0.6401
CLAHE	0.7629	0.6355

Table 2. Test performance of vanilla U-Net across preprocessing variants.

Figure 6 illustrates the training curve trends for validation loss and Dice score. These show consistent convergence, with no overfitting behavior observed across all three dataset variants. The smooth validation curves suggest the model generalizes well to unseen data.

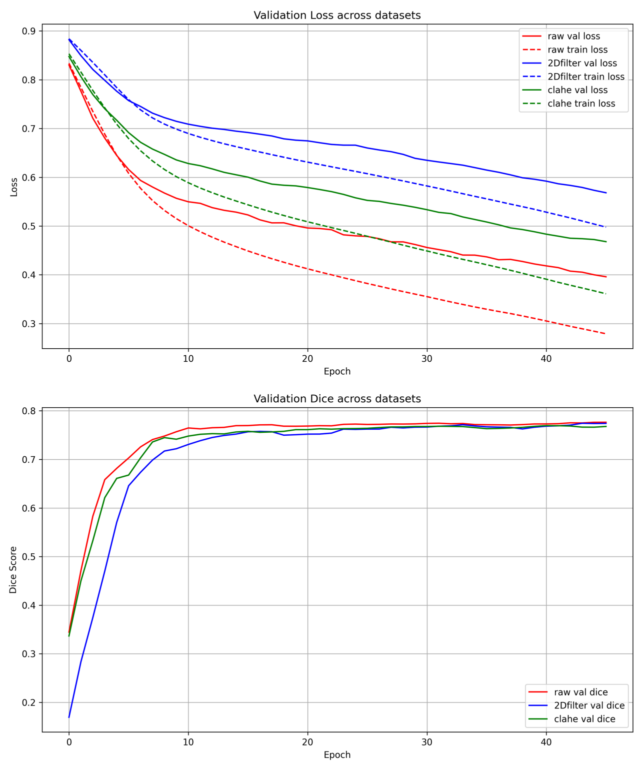

Figure 6. Vanilla U-Net performance: Top — validation loss across epochs; Bottom — validation Dice score. Raw (red), 2D filter (blue), and CLAHE (green).

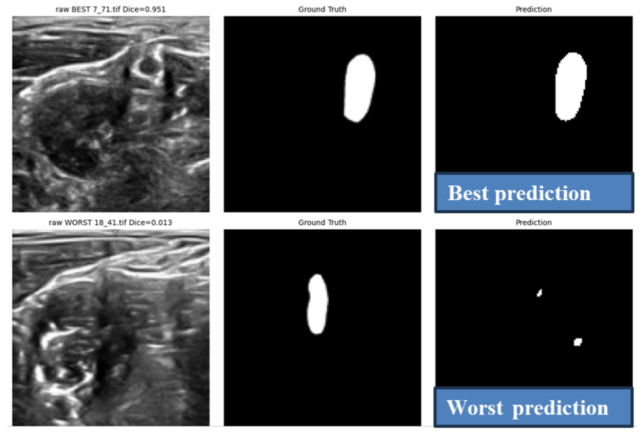

Figure 7. Qualitative results of vanilla U-Net on raw dataset. Left block: worst-case prediction; Right block: best-case prediction. Ground truth and prediction masks are shown alongside the original input.

We also examined qualitative results. In the worst case (Dice = 0.013), the model completely missed the nerve region. This was typically due to low contrast and ambiguous visual cues in the original image. Conversely, in the best case (Dice = 0.951), the segmentation closely matched the ground truth mask, with precise delineation of nerve boundaries. These examples highlight that while the model is capable of near-perfect performance, its predictions are sensitive to input quality (Fig. 7).

Inception U-Net Experiments

To do experiments on the Inception U-Net architecture, ultrasound images were resized to 64×64 pixels, normalized to [0,1], and augmented with random rotations ($\pm 15^\circ$) and flips to improve generalization. Training set is from patients 1-35, validation set is from patients 36 to 43, and the test set if from patients 44 to 47. The network was implemented in PyTorch with the following specifications: Convolutional weights were initialized using Xavier normal distribution to stabilize training [2], and for optimization we used Adam with a mini-batch size of 30. The learning rate is 0.001. We use β_1 of 0.9 and β_2 of 0.999 following [5]. The input of this network is the CLAHE-enhanced dataset. OneCycleLR is used to provide a structured approach to learning rate adjustment to allow for a faster convergence, improve generalization, and reduced hyperparameter tuning. We chose the peak to be at 1×10^{-2} and trained the network for 50 epochs on an NVIDIA GTX 1050 Ti GPU.

First, we started training the network without applying any learning rate scheduling strategy. The performance was monitored using Dice loss for training and validation and Dice coefficient for training and test sets. As can be seen in figure 8, the training loss continuously decreases over epochs which means that the model is successfully minimizing the loss on the training set. However, the validation loss initially decreases and then starts to plateau and also fluctuates. This discrepancy between training and validation loss suggests that the model may be overfitting, i.e. the model learns the training data too well and is failing to generalize to unseen validation samples. Figure 9 shows the training Dice coefficient steadily increases and reaches values above 0.85, showing the model is confident and accurate on the training data. On the other hand, the test Dice coefficient shows noisy behavior and stabilizes around 0.65 to 0.7, which also indicates overfitting observation.

To improve generalization and to address unstable test performance, we decided to use learning rate scheduling as mentioned before. The scheduler has a warm-up phase covering 30% of the training, and a sharp decrease toward the end. Figure 10 shows that the validation loss improved to a smoother curve and consistent decreasing and stabilizing at a lower value compared to before. Also, with lower divergence between the two curves, we can conclude that we have better generalization and less overfitting. Figure 11 shows that the training Dice coefficient improves steadily and reaches $\sim$ 0.9 by the end. The test Dice coefficient is now more stable and higher (around 0.8) with reduced variance across epochs. The IoU in this case is 0.6203.

Like Vanilla U-Net, we also examined qualitative results.

Figure 8. Inception U-Net performance: Loss function across epochs

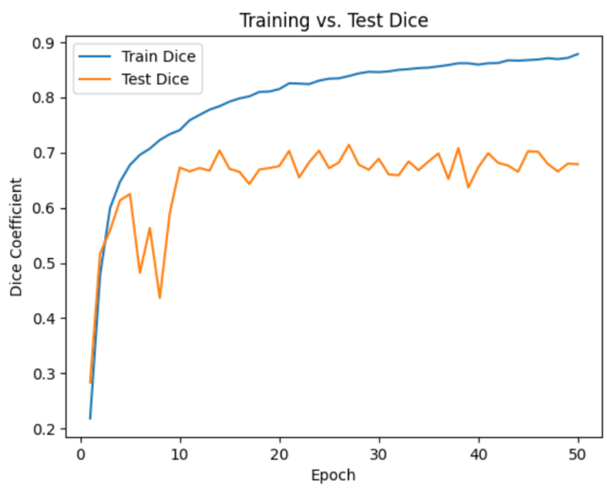

Figure 9. Inception U-Net performance: Dice score across epochs

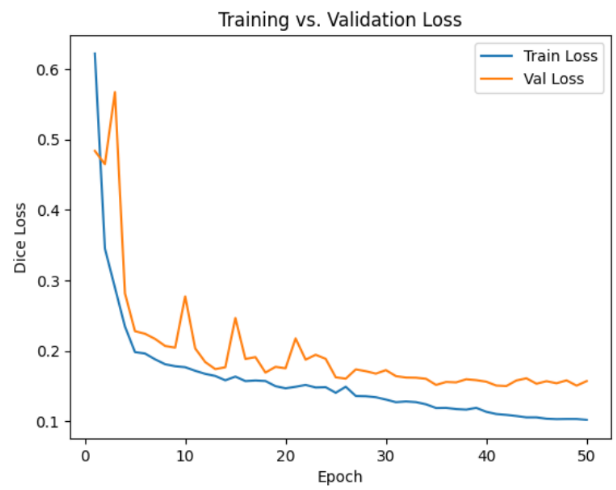

Figure 10. Inception U-Net performance after applying learning rate scheduler: Loss function across epochs

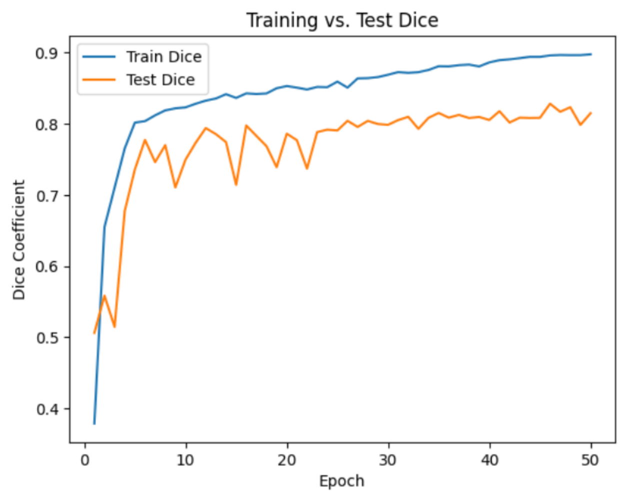

Figure 11. Inception U-Net performance after applying learning rate scheduler:Dice score across epochs

In the worst case (Dice=0.000), the model completely failed to detect nerve due to low contrast and ambiguous visual cues in the original image. On the other hand, in the best case (Dice = 0.927), the segmentation closely matched the ground truth mask, with precise delineation of the nerve boundaries (Fig. 12). Compared to baseline training, the Inception U-Net overall achieved better Dice score on the testing dataset and integration of the learning rate scheduler improved both convergence, speed, and generalization.

SegFormer Experiments

To further improve the results we started working with transformers. For the experiments with the SegFormer model, we used the `nvidia/segformer-b0` variant fine-tuned on binary nerve segmentation. Input images were resized to 512×512 pixels using Hugging Face's `SegformerFeatureExtractor`, and augmented with random flips and rotations to enhance generalizability. The training, validation, and test sets were split by patient IDs: 1–35 for training, 36–43 for validation, and 44–47 for testing. The model was trained using a hybrid loss combining Binary Cross-Entropy (BCEWithLogits) and Dice loss (foreground only), optimized with Adam and OneCycleLR (max LR = 3×10^{-5}). Mixed precision training (AMP) was enabled for efficiency.

We tracked training/validation losses and Dice coefficients per epoch. Figure 13 shows that both training and validation loss decreased consistently, and the gap between them remained small throughout training, suggesting minimal overfitting. The Dice coefficient curve in Figure 14 shows steady improvement in training accuracy and more stable test performance compared to Inception U-Net, with

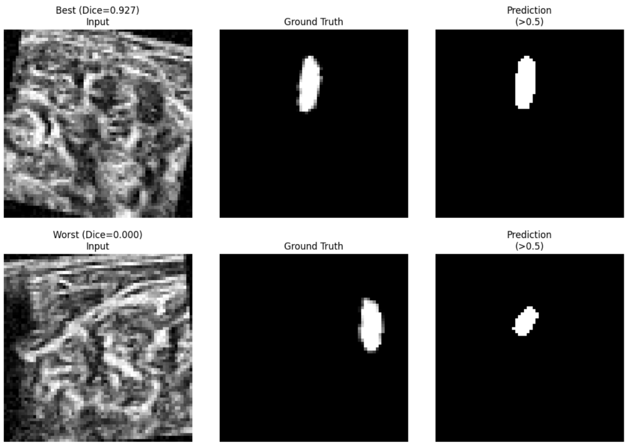

Figure 12. Qualitative results of Inception U-Net on augmented dataset with rotation. From right to left: ultrasound augmented image, ground truth mask, network's prediction.

final test Dice scores converging around 0.75–0.76 on average.

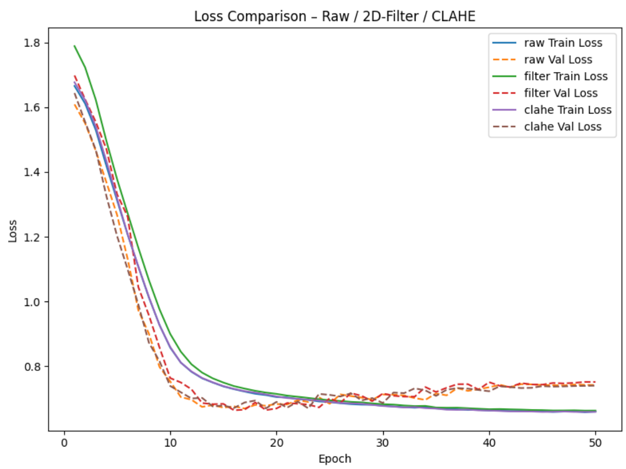

Figure 13. SegFormer performance: Combined BCE + Dice loss across epochs.

Preprocessing Comparisons. We compared various preprocessing pipelines on the test set, including no filtering (Raw), CLAHE enhancement, and 2D Gaussian filtering. Each method was applied alone or in combination. As shown in Table 3, the best performing combination was CLAHE on raw images, which achieved a Dice score of 0.759 with a corresponding loss of 0.924. Overall, CLAHE-based methods yielded more stable results across filtering conditions, achieving the best performance on Raw and CLAHE inputs, whereas 2D Filter have only marginal advantage having 2D Filter as input. This showed the CLAHE image pre-processing can improve generalizability of the derived model.

Qualitative Analysis. We also analyzed best- and worst-case predictions from different configurations. As shown in Figure 15, the best cases achieved Dice scores above 0.95, with high overlap between prediction and ground truth. The worst cases, on the other hand, have subpar performance on images without foreground vessels, leading to Dice scores of 0.0. These failures suggest future investigations in examples with blank ground truth masks where the model mistakenly predicted nerve structures.

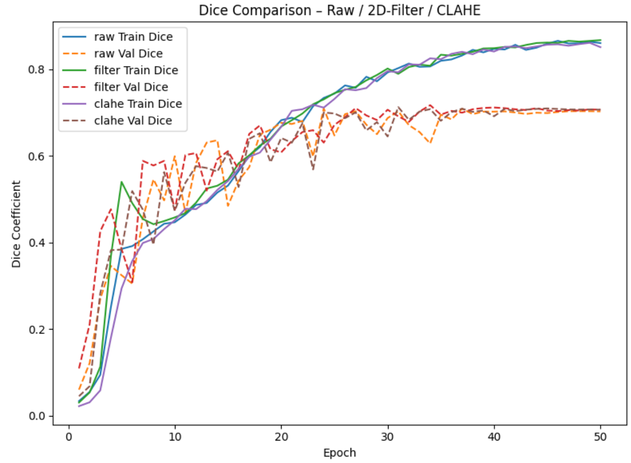

Figure 14. SegFormer performance: Dice coefficient across epochs.

Table 3. SegFormer Test Performance Across Preprocessing Methods

Preprocessing	Input	Loss	Dice
Raw	Raw	0.876	0.733
Filter	Raw	0.872	0.675
CLAHE	Raw	0.871	0.689
2D Filter	Raw	0.955	0.753
2D Filter	Filter	0.896	0.**730**
2D Filter	CLAHE	0.895	0.725
CLAHE	Raw	0.924	**0.759**
CLAHE	Filter	0.876	0.729
CLAHE	CLAHE	0.869	**0.730**

Overall, SegFormer outperformed Inception U-Net in terms of test stability, generalization, and Dice score, particularly when CLAHE preprocessing was used. Its transformer-based encoder and lightweight MLP decoder enabled both contextual awareness and edge precision, critical for segmenting fine nerve structures.

Discussion

This study explored three deep learning models—Vanilla U-Net, Inception U-Net, and SegFormer—for segmenting nerves in ultrasound images. Each model was evaluated under multiple preprocessing pipelines, helping us understand the strengths and limitations of both the architectures and the data preparation techniques.

Strengths. A key strength of this work is its breadth. We

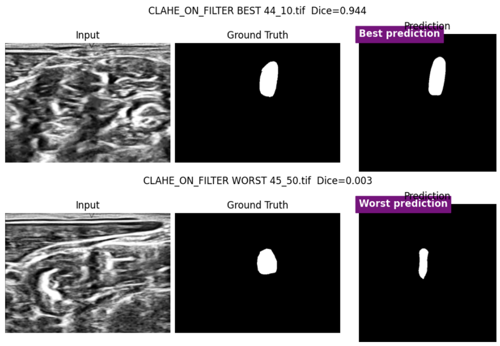

Figure 15. Qualitative SegFormer predictions: best and worst cases from various preprocessing pipelines. From left to right: input image, ground truth mask, model prediction.

compared fundamentally different models, from convolutional architectures to transformer-based networks, under controlled conditions. This allowed us to draw meaningful conclusions about their relative strengths. Our inclusion of learning rate scheduling, augmentation, and multiple loss functions showed how much implementation choices influence final performance. SegFormer stood out for its generalization ability despite being lightweight, while the Inception U-Net showed that architectural enhancements like multi-scale features can boost accuracy with proper training strategies. Preprocessing also played a crucial role — CLAHE in particular consistently improved results.

Weaknesses. There are several limitations to acknowledge. The dataset is relatively small, with some inconsistent or noisy labels, and includes images without nerves, which affected model reliability in edge cases. We only explored a single backbone variant for SegFormer due to computational limits, and we did not apply post-processing or ensemble methods, which could have improved boundary delineation or prediction stability. Additionally, some preprocessing techniques had marginal or even negative effects, suggesting that preprocessing needs to be better matched to model type and dataset properties.

Reflections. This project reinforced the idea that segmentation in ultrasound imaging is not just a model problem — it's a systems problem. Architecture, training strategy, data quality, and preprocessing all contribute to performance. Classical architectures like U-Net are still competitive when well trained, but modern transformer-based models offer more robustness, especially when image context matters. The gap between best- and worst-case predictions reminds us that even strong models can struggle in real-world scenarios, especially when ground truth is uncertain.

Conclusion

We presented a comparative study of three deep learning models for ultrasound nerve segmentation, using a diverse dataset with various preprocessing strategies. SegFormer with CLAHE preprocessing achieved the best test Dice score, while Inception U-Net showed gains when trained with learning rate scheduling. Vanilla U-Net remained a strong and consistent baseline. These findings highlight the value of transformer-based models in medical imaging, particularly when paired with effective preprocessing and training schemes.

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